

THE SPHERE-COVERING BOUND IN LEAN

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ABSTRACT. We formalize the classical sphere-covering lower bound on the covering number $K_q(n, R)$ in Lean 4, against Mathlib v4.34.1. The library proves that every code over $\mathbb{Z}/q\mathbb{Z}$ of length n which covers the space at Hamming radius R has at least $\lceil q^n/V \rceil$ words, where V is the size of a Hamming ball, and it checks eight numerical instances, including $K_7(9, 4) \geq 221$. The same archive determines several small covering numbers exactly, among them $K_2(7, 1) = 16$ by the binary Hamming code of length 7, and it exhibits a 12-word covering of $(\mathbb{F}_2)^6$ at radius 1. It does not prove matching upper bounds for the eight instances, does not decide whether those lower bounds improve the literature, and leaves $K_2(6, 1) \geq 11$ conditional on a search the kernel did not finish.

1. INTRODUCTION

The covering number $K_q(n, R)$ is the least cardinality of a code $C \subseteq A^n$, with $|A| = q$, such that every word of A^n lies at Hamming distance at most R from some word of C . The sphere-covering bound, already in Hamming and in the form recorded by Cohen, Honkala, Litsyn and Lobstein, says that if V is the number of words in a ball of radius R , then

$$(1) \quad q^n \leq K_q(n, R) \cdot V,$$

and therefore $K_q(n, R) \geq \lceil q^n/V \rceil$. The inequality is classical. What is not automatic is the arithmetic: for the cells in Table 1, V has up to six digits and q^n up to eight, and a hand evaluation of the ceiling is a place to slip.

This note reports a machine-checked proof of the bound and of eight of those ceilings, in the Lean 4 theorem prover [3] on top of Mathlib [4]. The source that was built is the tag `v0.2.0` of <https://github.com/thiagopatzdorf/Matematica>. A theorem below is a claim only when it is the name of a declaration in that tree and that declaration is imported by `CoveringLean.lean`. Two files in the tree, `A5b_Stress.lean` and `A6d_SearchHeavy.lean`, are not imported. Nothing in them is claimed.

2. DEFINITIONS

In the development `CoveringA2` the alphabet is $\mathbb{Z}/q\mathbb{Z}$ and a word is a function $\text{Fin } n \rightarrow \text{ZMod } q$. Distance is Mathlib's `hammingDist`. The ball of radius R about c is the finite set of words at distance at most R from c , and C covers at radius R when every word of the space lies in the ball of some element of C . The closed volume used for the numerical bound is

$$(2) \quad V(q, n, R) = \sum_{i=0}^R \binom{n}{i} (q-1)^i,$$

which is `CoveringA3.V`. The eight final statements do not define $K_q(n, R)$. Each says: every covering code has at least the stated number of words. That is the lower bound $K_q(n, R) \geq c$, not the determination of $K_q(n, R)$.

A second development, `CoveringA6`, uses the alphabet `Fin q` rather than $\mathbb{Z}/q\mathbb{Z}$, still with Mathlib's Hamming distance. There `IsKq n R k` is the proposition that some code of size k covers and

no code of size less than k does, so it is the equality $K_q(n, R) = k$. The two developments are not fused into one definition of K . The statements below name the development they come from.

3. THE BOUND

The argument in the library has four steps, each a theorem.

Theorem 1 (volume). *The number of words of Hamming weight at most R in $(\mathbb{Z}/q\mathbb{Z})^n$ equals $V(q, n, R)$. The generating function $\sum_x X^{\text{wt}(x)} = (1 + (q-1)X)^n$ is `CoveringA1.weight_genfun`; the ball at the origin is `CoveringA1.ball_zero_card`.*

Theorem 2 (translation). *For every centre c , the ball of radius R about c has the same cardinality as the ball about the origin (`CoveringA2.ball_card_indep`). The space itself has cardinality q^n (`CoveringA2.card_space`).*

Theorem 3 (sphere covering). *If C covers at radius R , then $q^n \leq |C| \cdot |B(0, R)|$ (`CoveringA2.sphere_covering`). If that volume does not divide q^n , the inequality is strict (`CoveringA2.no_perfect`). In the natural numbers, $\lceil q^n/V \rceil \leq |C|$ (`CoveringA2.ceil_bound`). Equality $|C| \cdot V = q^n$ forces the balls about the codewords to be pairwise disjoint (`CoveringA2.equality_iff_perfect`).*

Theorem 4 (identification). *`CoveringChain.ball_zero_eq_V` identifies the geometric ball of `CoveringA2`, cut out by `hammingDist`, with the closed sum (2). `CoveringChain.sphere_covering_formula` is Theorem 3 with that sum in place of the abstract volume.*

The eight numerical theorems apply the formula to a cell whose volume and ceiling have been evaluated. For each row of Table 1, `CoveringA3.cell...` packages the equalities $V(q, n, R) = v$ and $\lceil q^n/v \rceil = c$, together with the one-step obstruction $q^n \not\leq (c-1)v$. The obstruction says that the ceiling is sharp as a consequence of the sphere bound. It does not say that a code of size c exists.

q	n	R	V	$\lceil q^n/V \rceil$	declaration
5	7	2	365	215	SPH_K5_7_2_1b
4	10	4	20686	51	SPH_K4_10_4_1b
5	9	3	5989	327	SPH_K5_9_3_1b
5	10	4	62201	158	SPH_K5_10_4_1b
5	9	5	167269	12	SPH_K5_9_5_1b
5	9	4	38245	52	SPH_K5_9_4_1b
7	8	3	13153	439	SPH_K7_8_3_1b
7	9	4	182791	221	SPH_K7_9_4_1b

TABLE 1. Instances of the sphere-covering bound checked in `CoveringChain`. Each entry is a lower bound on every covering code, not a construction.

In particular, every covering of $(\mathbb{Z}/7\mathbb{Z})^9$ at radius 4 has at least 221 words. The library contains no theorem $K_7(9, 4) \leq 1351$, and it contains no list of 1351 words.

4. EXACT VALUES

On top of the lower bound, the library determines a few covering numbers completely.

Theorem 5. *If $R \geq n$, then one word covers and $K_q(n, R) = 1$ (`CoveringA4.K_eq_one_of_le`). For $n \geq 1$, $K_q(n, n-1) = q$: there is a covering of size q , and every covering has at least q words (`CoveringA4.K_pred_eq_q`).*

Theorem 6. *In `CoveringA6`, the following equalities hold: $K_2(2, 1) = 2$, $K_2(3, 1) = 2$, $K_2(4, 1) = 4$, $K_3(3, 2) = 3$, and $K_2(7, 1) = 16$.*

The last of these is the binary Hamming code of length 7. It is defined as the kernel of the parity-check matrix whose columns are $1, \dots, 7$. The kernel checks that the code has 16 words, that it covers at radius 1, that every ball of radius 1 has 8 words, that $|C| \cdot 8 = 2^7$, and that distinct codewords have disjoint balls. The sphere bound is therefore met with equality, and no code of size 15 covers.

Theorem 7. *The twelve words of $(\mathbb{F}_2)^6$ with integer encodings*

$$0, 1, 2, 15, 28, 23, 57, 58, 59, 39, 44, 52$$

form a code of cardinality 12 which covers at radius 1 (`CoveringA6.code12_card`, `CoveringA6.code12_covers`). Hence $K_2(6, 1) \leq 12$.

5. WHAT THE KERNEL ACCEPTED

The build that this note refers to is `lake build` of `CoveringLean` at Lean 4.34.1 and Mathlib tag `v4.34.1`. On 1 October 2026, on a virtual machine of type `e2-standard-8`, that build finished 8934 jobs in 34 seconds, with a peak resident set of 7.6 GB. The library imports no `sorry` and no `native_decide`. Declarations proved by `decide +kernel` were reduced in the kernel, not by a native evaluator. `#print axioms` on `CoveringChain.sphere_covering_formula` and on each of the eight `SPH_K*_lb` theorems reports at most `propext`, `Classical.choice` and `Quot.sound`.

Rebuilding requires `elan`, then `lake exe cache get` and `lake build` in the source tree. Running `decide +kernel` by hand on the universe of a large function type is a different computation: it asked for 17 to 24 GB and is not part of the build above.

6. WHAT IS NOT A THEOREM

The archive does not prove an upper bound for any row of Table 1. A lower bound of 221 is not a construction, and it is not a claim that 221 is the covering number.

The statement “no code of size at most 10 covers $(\mathbb{F}_2)^6$ at radius 1” is `CoveringA6.no_cover_2_6_1_le10`, and its hypothesis is

$$\text{refuted } 64 \ 7 \ \text{bm6} \ \text{nbr6} \ 10 \ 0 = \text{true}.$$

The file that would discharge that hypothesis by a kernel decision, `A6d_SearchHeavy.lean`, is not imported, does not build, and ran out of memory when it was tried. Until that equality is itself a theorem, $K_2(6, 1) \geq 11$ is not a theorem. The unconditional facts at this parameter are the sphere-covering method in general and the explicit upper bound $K_2(6, 1) \leq 12$.

The library does not compare Table 1 with the published tables of covering numbers, and it does not prove that any of the eight ceilings is new. The sphere bound is not new. The checked object is the proof.

REFERENCES

- [1] R. W. Hamming, *Error detecting and error correcting codes*, Bell System Technical Journal **29** (1950), 147–160.
- [2] G. Cohen, I. Honkala, S. Litsyn and A. Lobstein, *Covering Codes*, North-Holland, 1997.
- [3] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, Automated Deduction—CADE 28, Lecture Notes in Computer Science **12699**, Springer, 2021, 625–635.
- [4] The Mathlib Community, *Mathlib*, <https://github.com/leanprover-community/mathlib4>, release v4.34.1.

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